Experiment No.9

Experiment Title

Download, install and use Nmap with different options to scan open ports, perform OS fingerprinting, ping scan, TCP port scan, and UDP port scan.

Aim

To install and study the network scanning tool Nmap and perform different types of network scans such as Ping Scan, TCP Port Scan, UDP Port Scan, Open Port Scanning, and Operating System Detection.

Tools Required

Operating System: Microsoft Windows

Network Scanning Tool: Nmap

Internet Connection

Command Prompt / Command-line Interface

Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing. It helps administrators discover hosts on a network, identify open ports, detect services, and determine operating systems. Nmap works by sending specially crafted packets to the target host and analyzing the responses.

Types of Scans

1. Ping Scan – Checks whether a host is active.

2. TCP Port Scan – Detects open TCP ports and services like HTTP, FTP, SSH, HTTPS.

3. UDP Port Scan – Identifies UDP services like DNS, DHCP, SNMP.

4. OS Fingerprinting – Determines the operating system of the target system.

Installation Procedure

Step 1: Download Nmap from https://nmap.org/download.html

Step 2: Run the installer and follow instructions.

Install Nmap and Zenmap GUI.

Procedure

1. Open Command Prompt and verify installation using: nmap --version

2. Ping Scan: nmap -sn 192.168.1.1

3. Basic Port Scan: nmap -Pn 192.168.1.1

4. TCP Port Scan: nmap -sT -Pn 192.168.1.1

5. UDP Port Scan: nmap -sU -Pn 192.168.1.1

6. OS Detection: nmap -O -Pn 192.168.1.1

7. Service Version Detection: nmap -sV -Pn 192.168.1.1

Observation Table :

Scan Type	**Command**	**Purpose**
Ping Scan	nmap -sn target	Detect active host
Basic Scan	nmap -Pn target	Detect open ports
TCP Scan	nmap -sT -Pn target	Scan TCP ports
UDP Scan	nmap -sU -Pn target	Scan UDP ports
OS Detection	nmap -O -Pn target	Detect operating system
Service Detection	nmap -sV -Pn target	Detect service versions

Result

The Nmap tool was successfully installed and used to perform host discovery, open port scanning, TCP port scanning, UDP port scanning, and operating system detection.

Conclusion

Nmap is a powerful network scanning and security auditing tool widely used in network administration and cybersecurity testing.